Web3's HTML equivalent are CAIPs, chain-agnostic data formats specified by the Chain Agnostic Standards Alliance (CASA). All wallets using WalletConnect embrace these standards and they allow web3 to remain open. The emergence of proprietary alternatives come at the expense of privacy and censorship resistance - it is important for dapps and users to demand them to use open formats so that web3 remains open.

Since its inception, the vision has been that the WalletConnect Network should be decentralized. Not having a single entity control the Network improves its resilience and censorship resistance. A free centralized service results in "payment" in other ways - through loss of privacy because users' data is being sold, or loss of censorship resistance because you can be offboarded. A decentralized network supported by the ecosystem is the only solution that truly meets the Web3 ethos.

The Network's design emphasizes interoperability, connectivity, and composability, creating a user experience (UX) platform that is open and permissionless. This approach is essential for realizing the vision of the "new internet"—a decentralized and user-owned ecosystem.

4 Technology Overview

The WalletConnect Network's services are more akin to classic web2 offchain infrastructure rather than blockchain: it provides webservices that require state and their operation is subject to low latency and high throughput requirements.

There was no off-the-shelf permissionless system available that was proven at scale and that met the latency requirements (see 4.2 below) which is why WalletConnect ended up marrying concepts of the blockchain and database worlds when building the Network.

4.1 Core Technology

The core technology for the WalletConnect Network is a permissionless rendezvous-hashing based database. Rendezvous hashing is a concept that all modern databases including Cassandra, DynamoDB, MongoDB, etc are based upon. It's proven for global scale and it is decentralized by design as its lack of a central point of failure allows it to scale. However, none of these existing databases are permissionless as this is not a design goal they needed to achieve.

100% of WalletConnect's production traffic has been served by the network since early 2024 with Service Nodes operated by Reown. In recent months Reown has brought additional entities into the permissioned federation of Service Node operations.

As such, WalletConnect Network also constitutes a new primitive in the decentralized application builders' toolbox: a cost efficient/low latency/high throughput key value database serving WalletConnect which is one of web3's most used applications.